

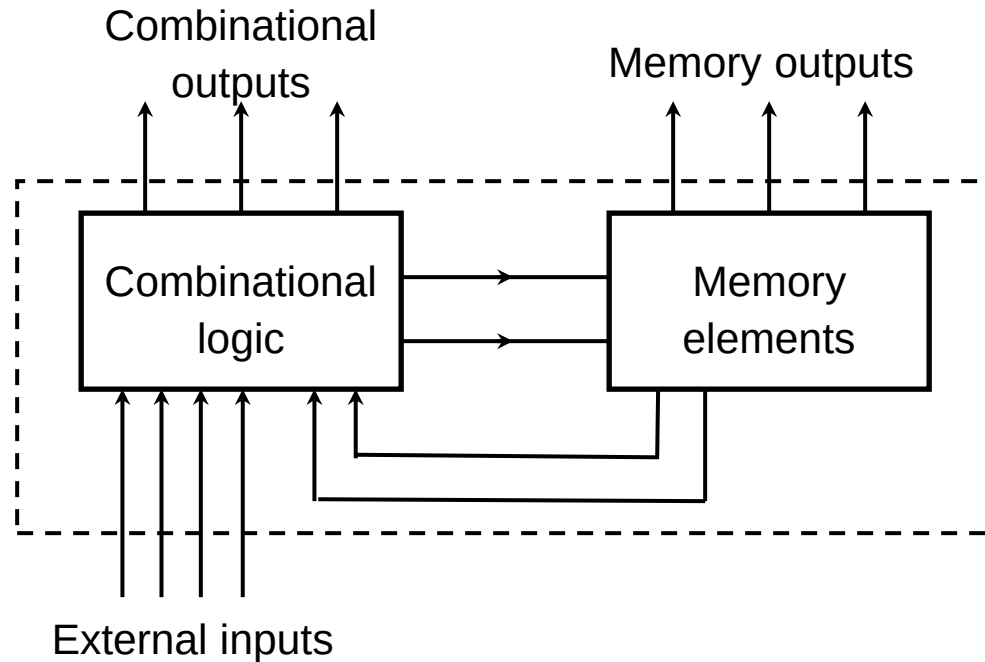
CSE 260

DIGITAL LOGIC DESIGN

Sequential Logic, RS Flip-Flop,
D Flip-Flop, JK Flip-Flop, T Flip-Flop
BRAC University

Introduction

- A **sequential circuit** consists of a *feedback path*, and employs some *memory elements*.



Sequential circuit = Combinational logic + Memory Elements

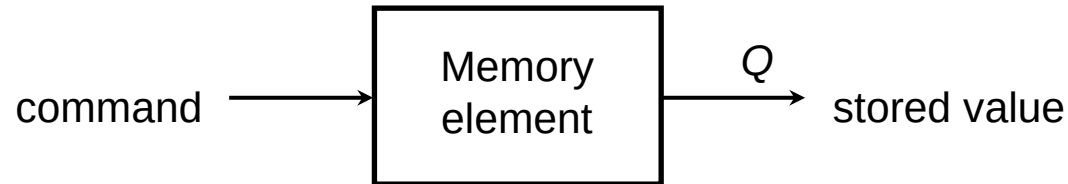
output = external input + present state of memory element

Introduction

- There are two types of sequential circuits:
 - ❖ *synchronous*: outputs change only at specific time (i.e. with clock input)
 - ❖ *asynchronous*: outputs change at any time (i.e. without clock input)
- *Multivibrator*: a class of sequential circuits. They can be:
 - ❖ *bistable* (2 stable states)
 - ❖ *monostable* or *one-shot* (1 stable state)
 - ❖ *astable* (no stable state)
- Bistable logic devices: *flip-flops*.
- Flip-flops differ in the method used for changing their state.

Memory Elements

- **Memory element**: a device which can remember value indefinitely, or change value on command from its inputs.



- **Characteristic table**:

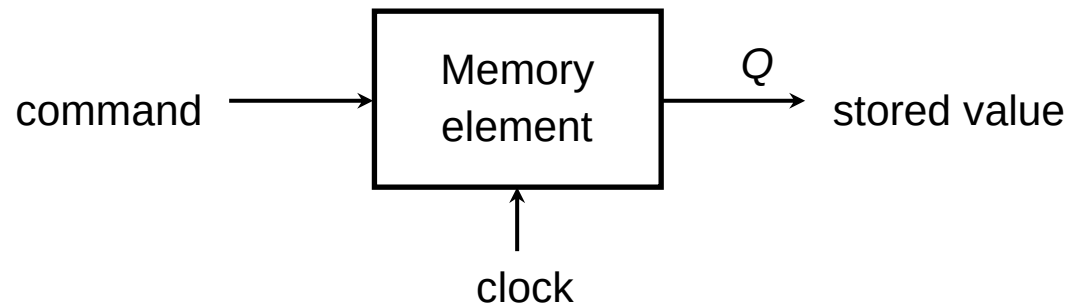
Command (at time t)	$Q(t)$	$Q(t+1)$
Set	X	1
Reset	X	0
Memorise / No Change	0	0
	1	1

$Q(t)$: current state

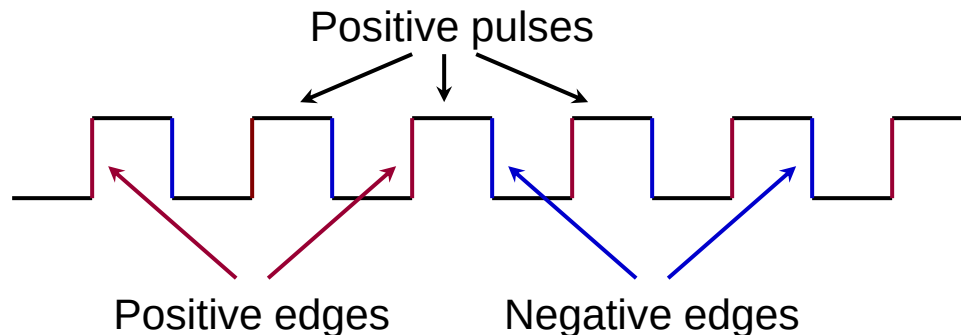
$Q(t+1)$ or Q^+ : next state

Memory Elements

- Memory element with clock. Flip-flops are memory elements that change state on clock signals.



- Clock is usually a square wave.



Types of tables in sequential circuit

- Characteristic table
- Criteria Table
- State Table
- Excitation table

$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate

J	K	$Q(t+1)$	Comments
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q(t)'$	Toggle

Present state		Input x	Next state		Flip-flop inputs			
A	B		A^+	B^+	J_A	K_A	J_B	K_B
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	X
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	X	1

Present State		Input x	Next State		Output y
A	B		A^+	B^+	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

S-R Flip-Flop

- *Complementary* outputs: Q and Q' .
- When Q is HIGH, the FF is in *SET* state.
- When Q is LOW, the FF is in *RESET* state.
- For *active-HIGH input S-R FF* (also known as NOR gate FF),

$R=\text{HIGH}$ (and $S=\text{LOW}$) $\Rightarrow$ RESET state

$S=\text{HIGH}$ (and $R=\text{LOW}$) $\Rightarrow$ SET state

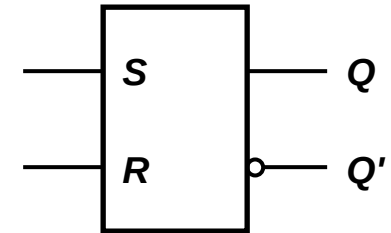
both inputs LOW $\Rightarrow$ no change

both inputs HIGH $\Rightarrow$ Q and Q' both LOW (invalid)!

S-R FF

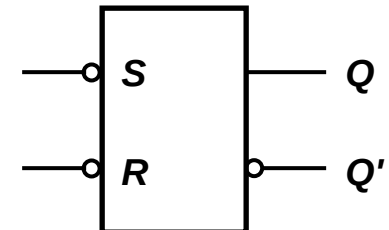
- Characteristics table for **active-high** input S-R FF:

S	R	Q	Q'	
0	0	NC	NC	No change. FF remained in present state.
1	0	1	0	FF SET.
0	1	0	1	FF RESET.
1	1	0	0	Invalid condition.



- Characteristics table for **active-low** input S'-R' FF:

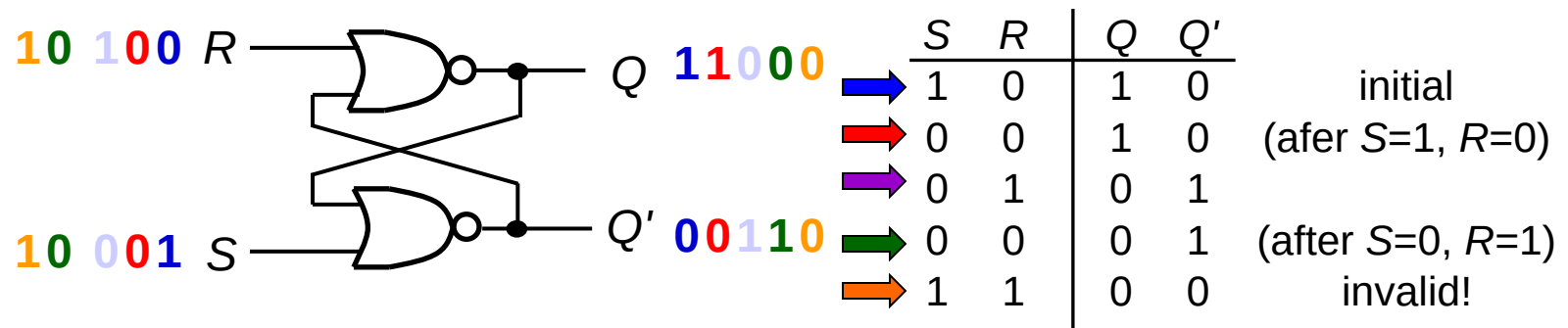
S'	R'	Q	Q'	
1	1	NC	NC	No change. FF remained in present state.
0	1	1	0	FF SET.
1	0	0	1	FF RESET.
0	0	1	1	Invalid condition.



Use this mostly

Notice the difference

S-R FF: Active-HIGH input S-R FF



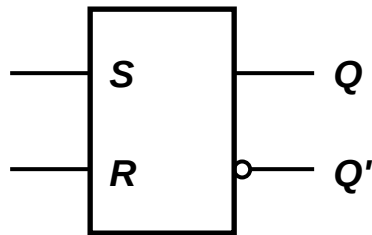
NOR truth table

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

Presence of '1' in input, leads to '0' in output

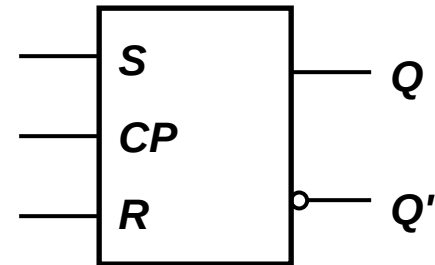
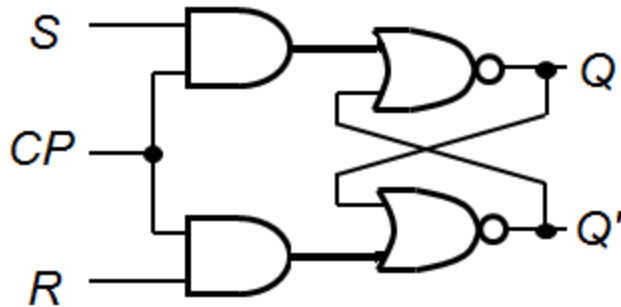
S-R FF

R	S	Operation	Q(t)	Q(t+1)	Q'(t)	Q'(t+1)
0	0	No Change	0	0	1	1
			1	1	0	0
0	1	Set	x	1	x	0
1	0	Reset	x	0	x	1
1	1	Invalid	-	-	-	-



Clocked S-R FF

- S-R FF + Clock Pulse (CP) and 2 NAND gates → Clocked *S-R FF*.



Clocked S-R FF

- Outputs change (if necessary) only when **CP is HIGH**.
- Under what condition does the invalid state occur?
- Characteristic table: Characteristic Eqⁿ:

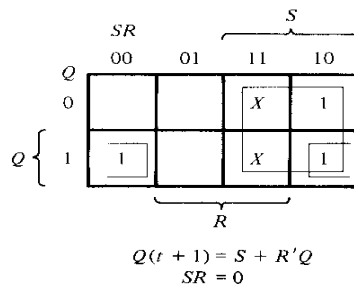
CP=1

$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate

$$Q(t+1) = S + R'.Q$$

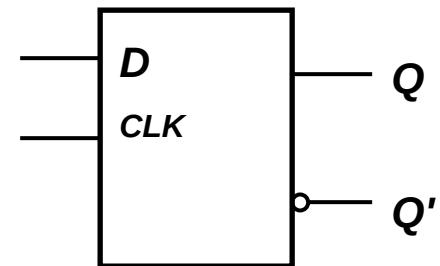
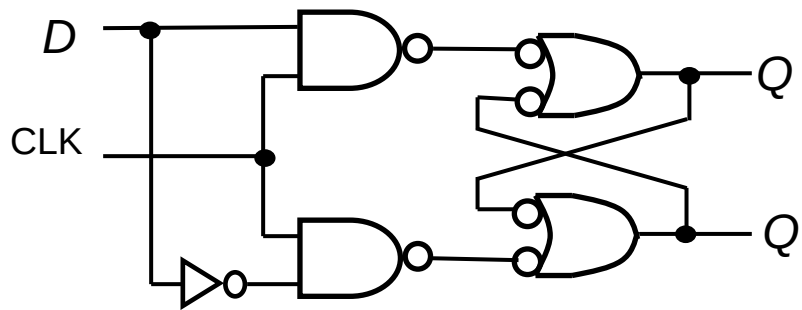
$$S.R = 0$$

S	R	$Q(t+1)$
0	0	$Q(t)$ No change
0	1	0 Reset
1	0	1 Set
1	1	indeterminate



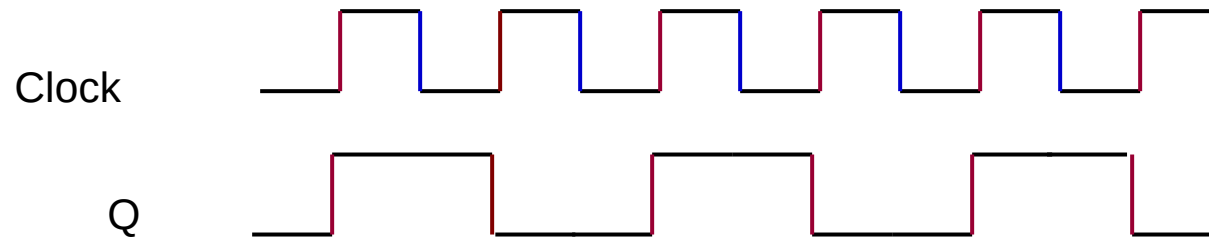
Clocked D Flip-Flop

- Make R input equal to S' $\rightarrow$ *D FF*.
- *D FF* eliminates the undesirable condition of invalid state in the *S-R FF*.



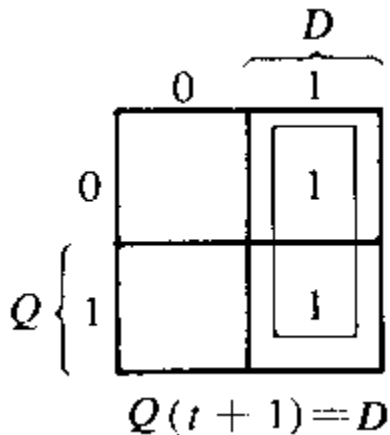
D Flip-flop Characteristic table

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1



Clocked D Flip-Flop

- When CLK is HIGH,
 - ❖ $D=\text{HIGH} \rightarrow \text{FF is SET}$
 - ❖ $D=\text{LOW} \rightarrow \text{FF is RESET}$
- Hence when CLK is HIGH, Q 'follows' the D (data) input.
- Characteristic table:



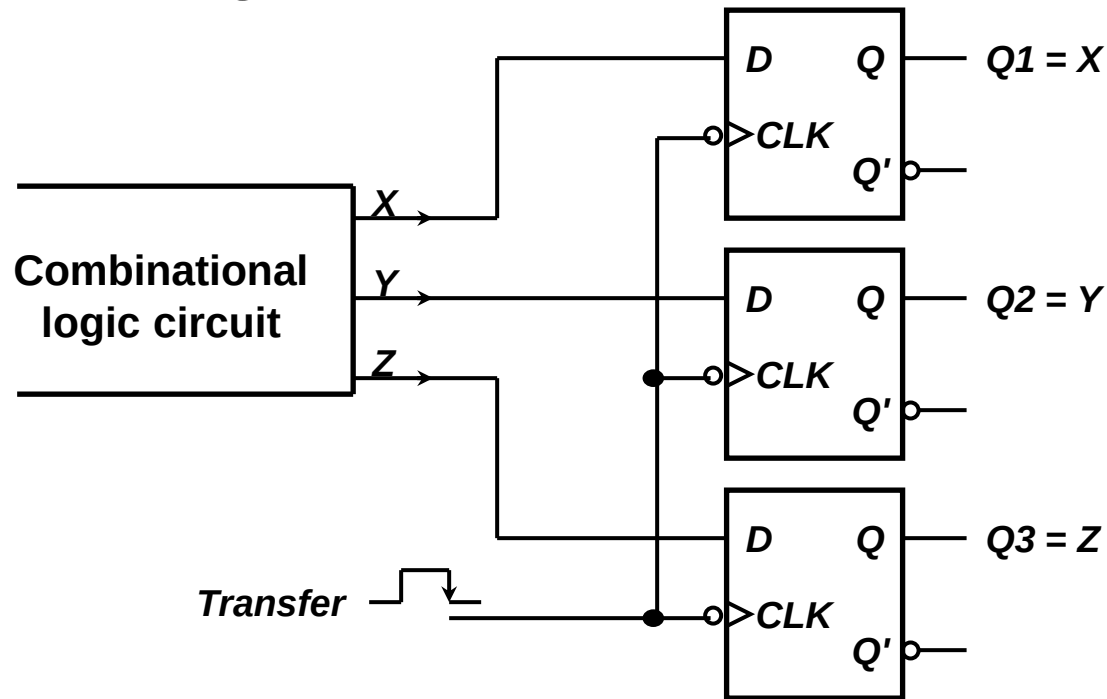
CLK	D	$Q(t+1)$	
1	0	0	Reset
1	1	1	Set
0	X	$Q(t)$	No change

When $CLK=1$, $Q(t+1) = D$

D Flip-flop

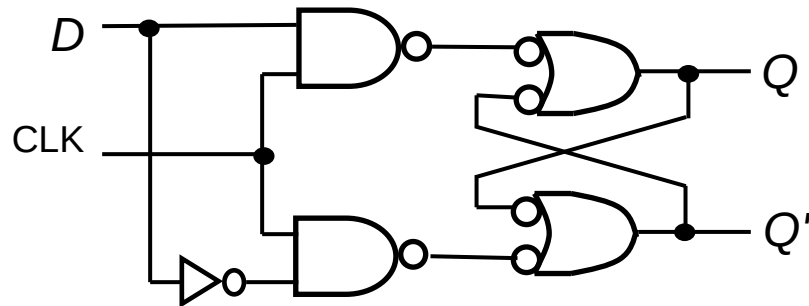
- Application: *Parallel data transfer*.

To transfer logic-circuit outputs X , Y , Z to flip-flops Q_1 , Q_2 and Q_3 for storage.



Try it yourself

- Design a D FF using RS FF



J-K Flip-flop

- J-K flip-flop: Q and Q' are fed back to the NAND gates.
- No invalid state.
- Include a *toggle* state.
 - ❖ $J=\text{HIGH}$ (and $K=\text{LOW}$) $\Rightarrow$ SET state
 - ❖ $K=\text{HIGH}$ (and $J=\text{LOW}$) $\Rightarrow$ RESET state
 - ❖ both inputs LOW $\Rightarrow$ no change
 - ❖ both inputs HIGH $\Rightarrow$ toggle

J-K FF

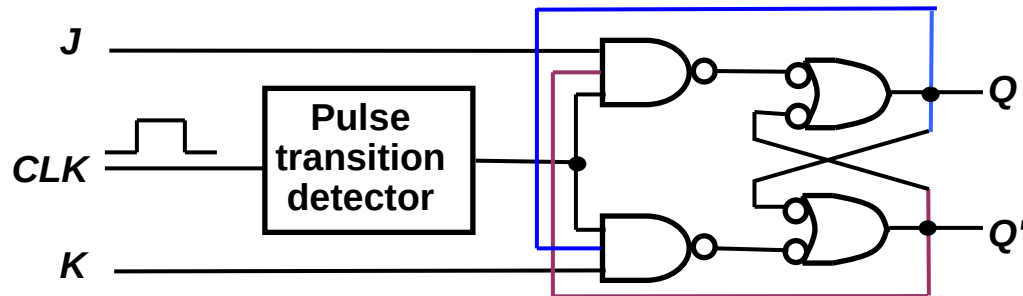
SET

RESET

Clock	J	K	Operation	Q(t)	Q(t+1)	Q'(t)	Q'(t+1)
↑	0	0	No Change	0	0	1	1
				1	1	0	0
↑	1	0	Set	x	1	x	0
↑	0	1	Reset	x	0	x	1
↑	1	1	Toggle	1	0	0	1

J-K Flip-flop

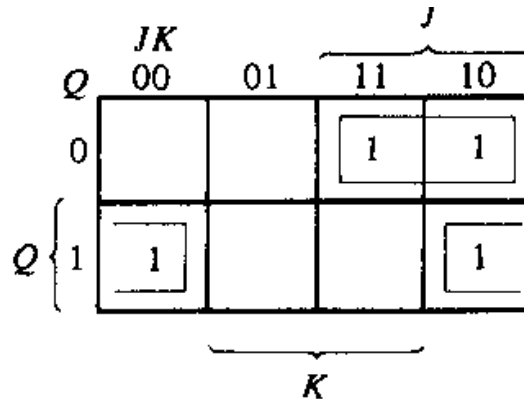
- J-K flip-flop.



J	K	CLK	Q(t+1)	Comments
0	0	↑	Q(t)	No change
0	1	↑	0	Reset
1	0	↑	1	Set
1	1	↑	Q(t)'	Toggle

- Characteristic table.

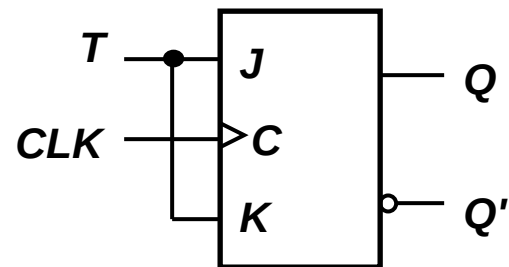
Q	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



$$Q(t+1) = J.Q' + K'.Q$$

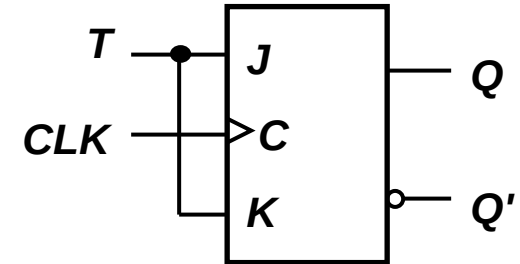
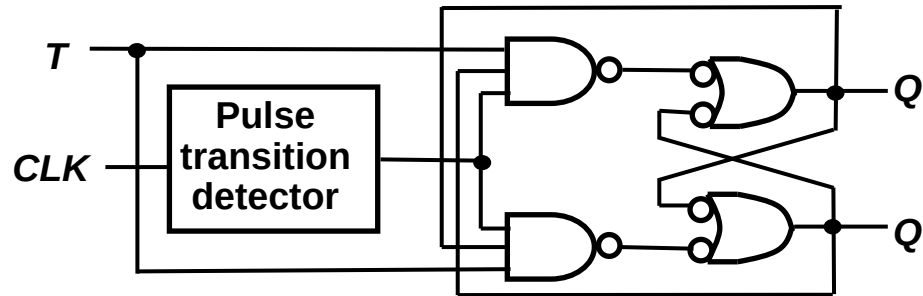
T Flip-flop

T	Operation	Q(t)	Q(t+1)
0	No change	0	0
		1	1
1	Toggle	0	1
		1	0



T Flip-flop

- **T flip-flop**: single-input version of the J-K flip flop, formed by tying both inputs together.



- Characteristic table.

T	CLK	$Q(t+1)$	Comments
0	↑	$Q(t)$	No change
1	↑	$Q(t)'$	Toggle

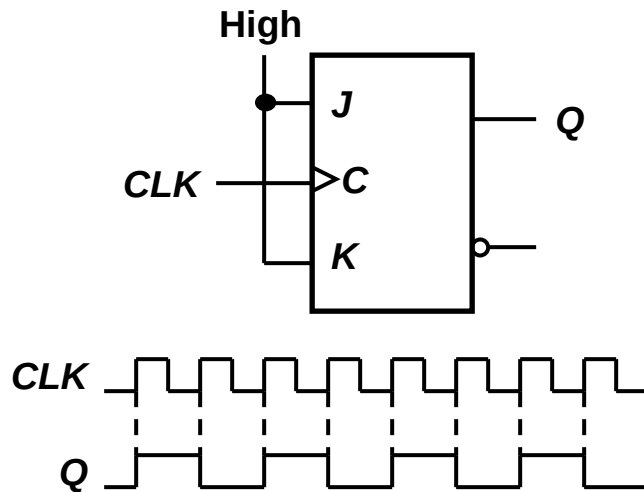
Q	T	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

		T
		0 1
Q	0	□ 1
1	1	□ □

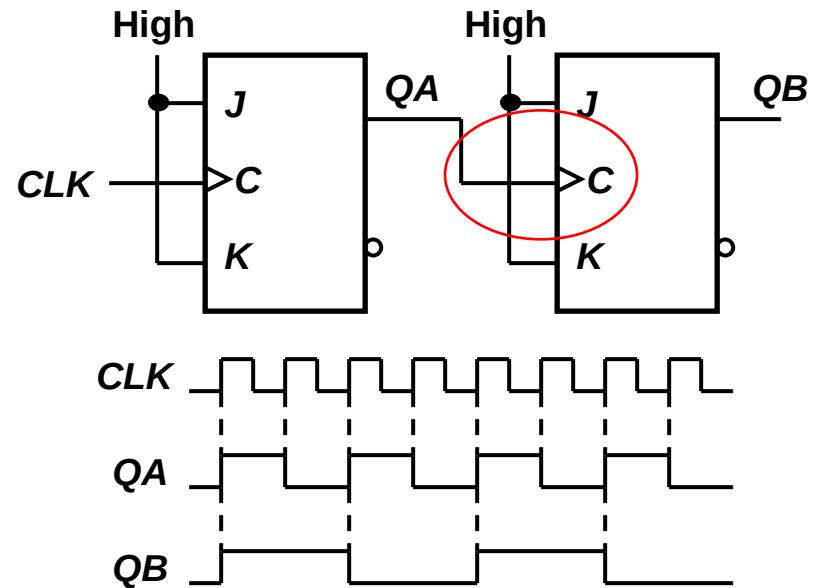
$$Q(t+1) = T.Q' + T'.Q$$

T Flip-flop

- Application: *Frequency division*.



Divide clock frequency by 2.

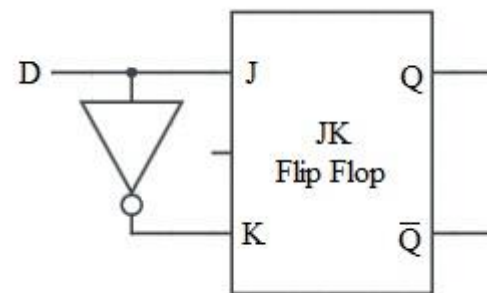
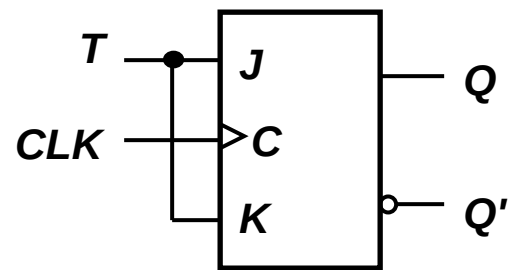


Divide clock frequency by 4.

- Application: *Counter*

Try it yourself

- Design a T FF using JK FF
- Design a D FF using JK FF

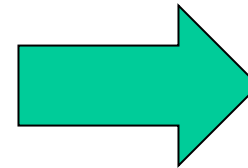


J	K	CLK	$Q(t+1)$	Comments
0	0	$\uparrow$	$Q(t)$	No change
0	1	$\uparrow$	0	Reset
1	0	$\uparrow$	1	Set
1	1	$\uparrow$	$Q(t)'$	Toggle

How to build excitation table:

Example: JK Flip-Flop

Q	Q+	Actually what happens		Final Combined Result	
		J	K	J	K
0	0	0	1	0	x
		0	0		
0	1	1	0	1	x
		1	1		
1	0	0	1	x	1
		1	1		
1	1	0	0	x	0
		1	0		



Q	Q^*	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

JK Flip-flop

Flip-flop Excitation Tables

- Excitation tables : it give transition characteristic between current condition and next condition to determine flip-flop input

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-flop

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

SR Flip-flop

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

D Flip-flop

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

T Flip-flop